

Jit Malla [0.25em]

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Research Profile

Computer Science researcher and **Ph.D. Research Scholar in Computer Science and Engineering at IIT Kharagpur**, with research experience spanning **computer vision, multimodal learning, vision-language models (VLMs), deep learning, and graph-based visual reasoning**. My research focuses on connecting visual perception with structured reasoning by combining **scene graphs, vision-language models, large language models, and verification mechanisms**. Current interests include the **robustness, reliability, interpretability, hallucination, and reasoning capabilities of multimodal AI systems**, particularly for visual grounding, spatial reasoning, and intelligent systems operating in complex real-world environments.

Research Experience

Ph.D. Research — IIT Kharagpur 2026 – Present *Computer Vision, Multimodal AI, Vision-Language Reasoning and Verification*

- Investigating **multimodal AI systems and vision-language models** with an emphasis on visual reasoning, grounding, reliability, and systematic verification of model-generated responses.
- Exploring approaches that combine **visual perception, structured representations, agentic reasoning, and verification** to improve the reliability of multimodal reasoning systems.
- Studying the limitations of current VLM-based systems, including **hallucination, reasoning errors, inconsistent predictions, and propagation of errors across multimodal pipelines**.
- Developing research directions inspired by recent work on **multimodal verification and vision-language agents**, including the integration of verification-oriented reasoning with agent-based visual understanding.

M.Sc. Research Project — Indian Association for the Cultivation of Science (IACS), Kolkata 2024 – 2026

Scene Graph Based Spatial Reasoning and Visual Grounding in Laboratory Environments using Vision-Language Models

- Developed a **scene graph generation pipeline** for chemistry laboratory images using vision-language models under both image-only and image-with-bounding-box settings.
- Conducted a systematic investigation showing that **bounding-box supervision improves object identification and spatial reasoning**, with measurable improvements in recall and F1-score.
- Generated structured **scene graphs containing objects, attributes, and spatial relations**, followed by post-processing to improve geometric consistency and relational accuracy.
- Leveraged **vision-language models and large language models** for spatial reasoning and question answering over generated scene graphs.
- Investigated **verification mechanisms** for improving the logical consistency of model-generated spatial reasoning and reducing errors in multimodal reasoning pipelines.
- Studied practical challenges involving **hallucination, error propagation, visual grounding, and reliability** in VLM-based systems, motivating further research toward layered and real-time 3D scene graph representations.

B.Sc. Project — Ramakrishna Mission Residential College, Narendrapur 2023 – 2024

A Wavelet-Based Approach for Authenticating Medical Images and Extracting Patient Information Guide: Bibek Ranjan Ghosh, Assistant Professor

- Developed a secure medical image authentication framework using **wavelet transforms** for embedding and extracting patient metadata.
- Designed the framework to preserve **diagnostic image quality** while maintaining data integrity.
- Evaluated robustness against **compression, noise, and common image-processing distortions**.

Current Research Directions

- **Multimodal AI Reliability and Verification:** Studying mechanisms for verifying visual-language model outputs and detecting reasoning inconsistencies.

- **Vision-Language Agents:** Exploring agent-based approaches for visual understanding, reasoning, and task decomposition.
- **Scene Graphs and Structured Visual Reasoning:** Using graph-based representations to bridge visual perception and symbolic reasoning.
- **Visual Grounding and Spatial Reasoning:** Investigating object localization, spatial relations, and reasoning over complex visual environments.
- **Multimodal Hallucination and Error Propagation:** Understanding failure modes in VLM-based multimodal pipelines and developing methods for more reliable reasoning.

Technical Skills

Programming: Python, C, C++, Java

Machine Learning Deep Learning: PyTorch, Transformer-based Models, Convolutional Neural Networks (CNNs), Feature Extraction, Representation Learning, Deep Learning

Computer Vision: OpenCV, YOLO, Visual Grounding, Image Understanding, Object Detection, Spatial Reasoning

Multimodal AI Reasoning: Vision-Language Models (VLMs), Large Language Models (LLMs), Multimodal Reasoning, Scene Graph Generation, Visual Question Answering, Model Verification, Agent-based Reasoning

Graph-Based Vision: Scene Graphs, Object-Attribute Relationship Modeling, Spatial Relationship Inference, Graph-based Knowledge Representation, Graph Visualization

Tools Frameworks: Hugging Face Transformers, NumPy, PyDot, Visual Studio Code, Linux/Unix

Documentation: LaTeX

Education

Ph.D. in Computer Science and Engineering 2026 – Present **Indian Institute of Technology Kharagpur (IIT Kharagpur)**

M.Sc. in Computer Science 2024 – 2026 **Indian Association for the Cultivation of Science (IACS), Kolkata** Final CGPA: **8.15/10**

B.Sc. in Computer Science 2021 – 2024 **Ramakrishna Mission Residential College, Narendrapur**

Academic Qualifications

- Qualified **UGC NET** — **June 2026**.
- Qualified **GATE 2026** in Computer Science and Information Technology.

Relevant Coursework

Computer Science: Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Theory of Computation, Digital Image Processing, Algorithm Analysis, Software Engineering, Computer Networks, Artificial Intelligence, Machine Learning, Artificial Intelligence Data Science, Computer Architecture, Discrete Mathematics

Mathematics Statistics: Linear Algebra, Ordinary Differential Equations, Probability Theory, Statistical Inference, Sampling Methods, Descriptive Statistics, Numerical Analysis